

Credit Risk Analysis

Created by Doan My Van

August, 2025



AGENDA OVERVIEW

01

BACKGROUND

02

PROBLEM STATEMENT & OBJECTIVE

03

ANALYSIS FRAMEWORK

04

METHODOLOGY

05

ANALYSIS

06

KEY INSIGHTS

07

RECOMMENDATION





The company is a financial technology (Fintech) platform that connects investors (lenders) with borrowers, enabling peer-to-peer lending outside the traditional banking system. The company facilitates loans where investors provide capital and borrowers repay with interest, generating profit for both investors and the company.

Background



Problem Statement

- ❑ With the rise in loan demand, assessing borrower risk and loan performance has become increasingly important.
- ❑ The dataset, spanning 2007–2014 and including the 2008 financial crisis, poses potential risk implications.
- ❑ Analyzing this data can uncover repayment behaviors, default trends, and key factors influencing credit risk

Objective ■

- ❑ Identify key factors influencing borrower behavior and repayment capacity
- ❑ Distinguish between good and bad loans through data analysis
- ❑ Propose strategies to enhance lending decisions and reduce defaults

■ METHODOLOGY

Loan Classification:

❑ Good loan:

- Fully Paid
- Charge off, default, late (recovery rate > 70%)
- Current (Loan seasoning >40%)

❑ Bad loan:

- Charge off, default, late (recovery rate <40%)

❑ Neutral: Else

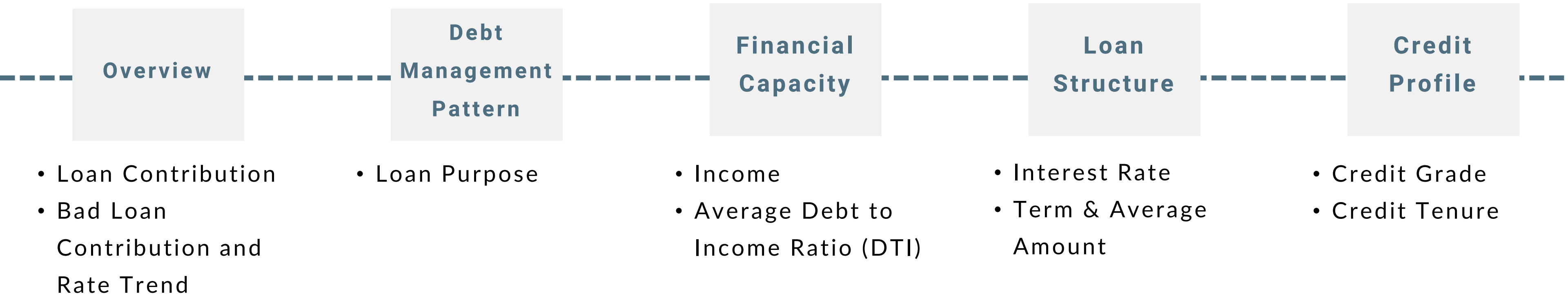
Key dimensions:

- ❑ Credit profiles
- ❑ Debt management pattern
- ❑ Income and financial stability
- ❑ Loan structure and terms





ANALYSIS FRAMEWORK



Low-performing loans increased despite Good loans making up the majority of the portfolio

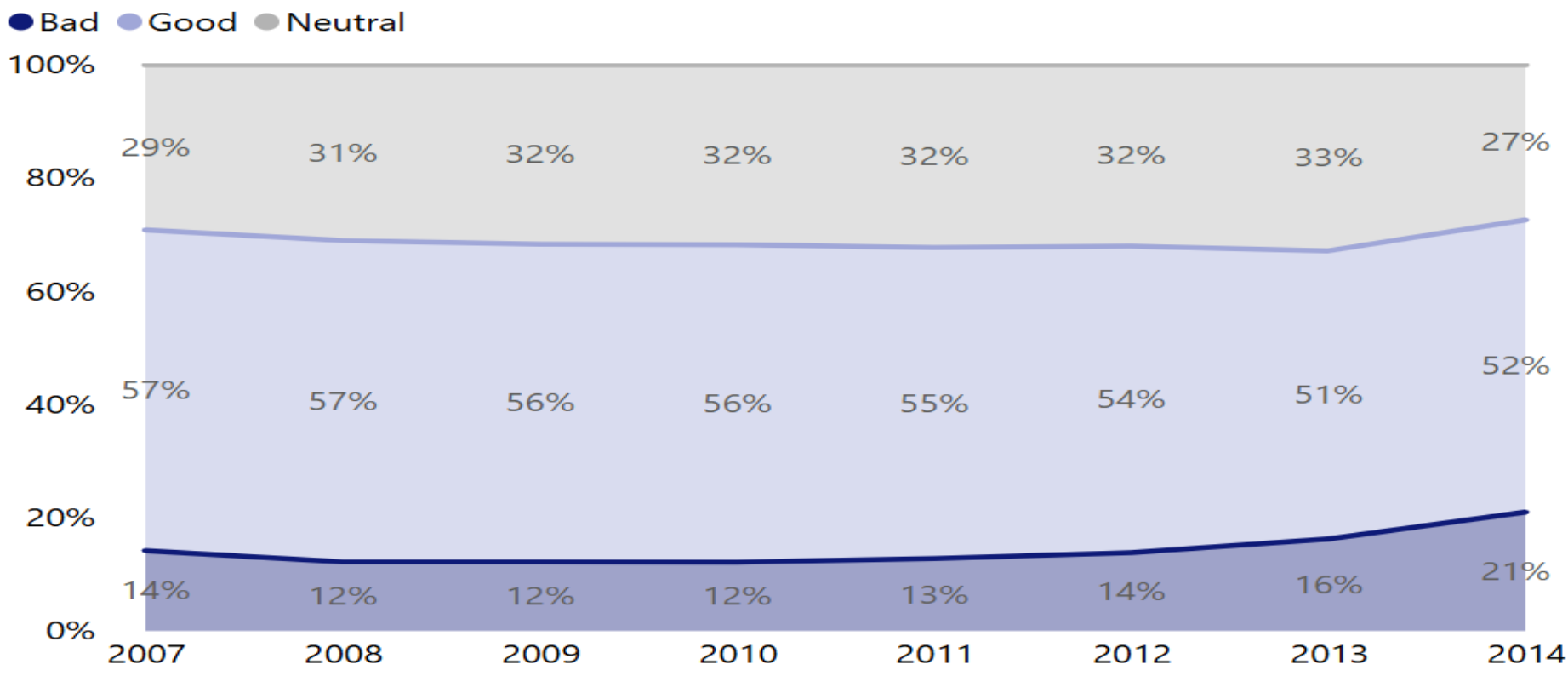
Most of the loan was contributed by Good Loan ~71%, followed by Neutral ~19% and nearly 10% Bad Loan

Loan Contribution by Loan Classification



Bad loans rose by 7% from 2007 to 2014—especially in the later years—while Good loans slightly declined

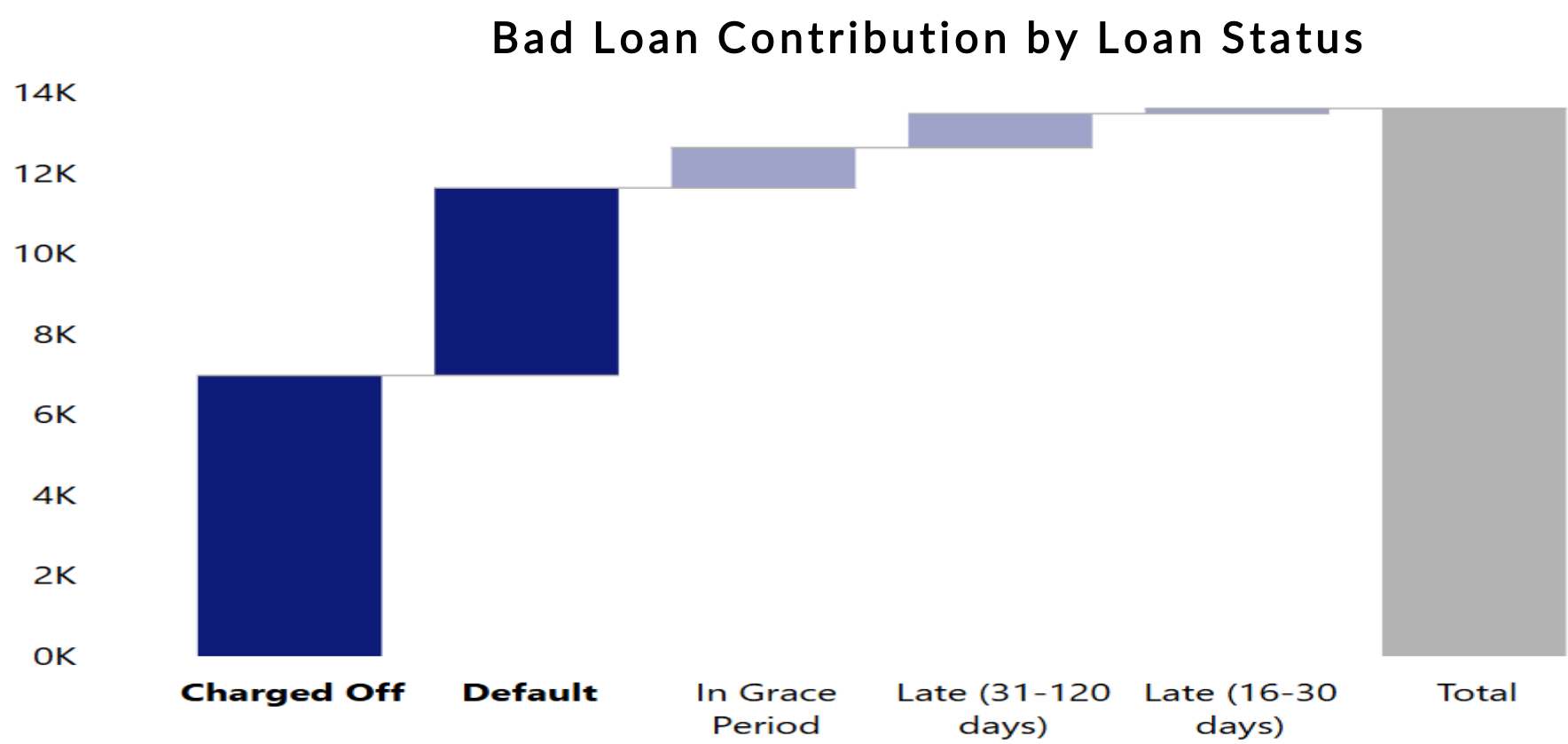
Average Recovery Trend by Loan Classification



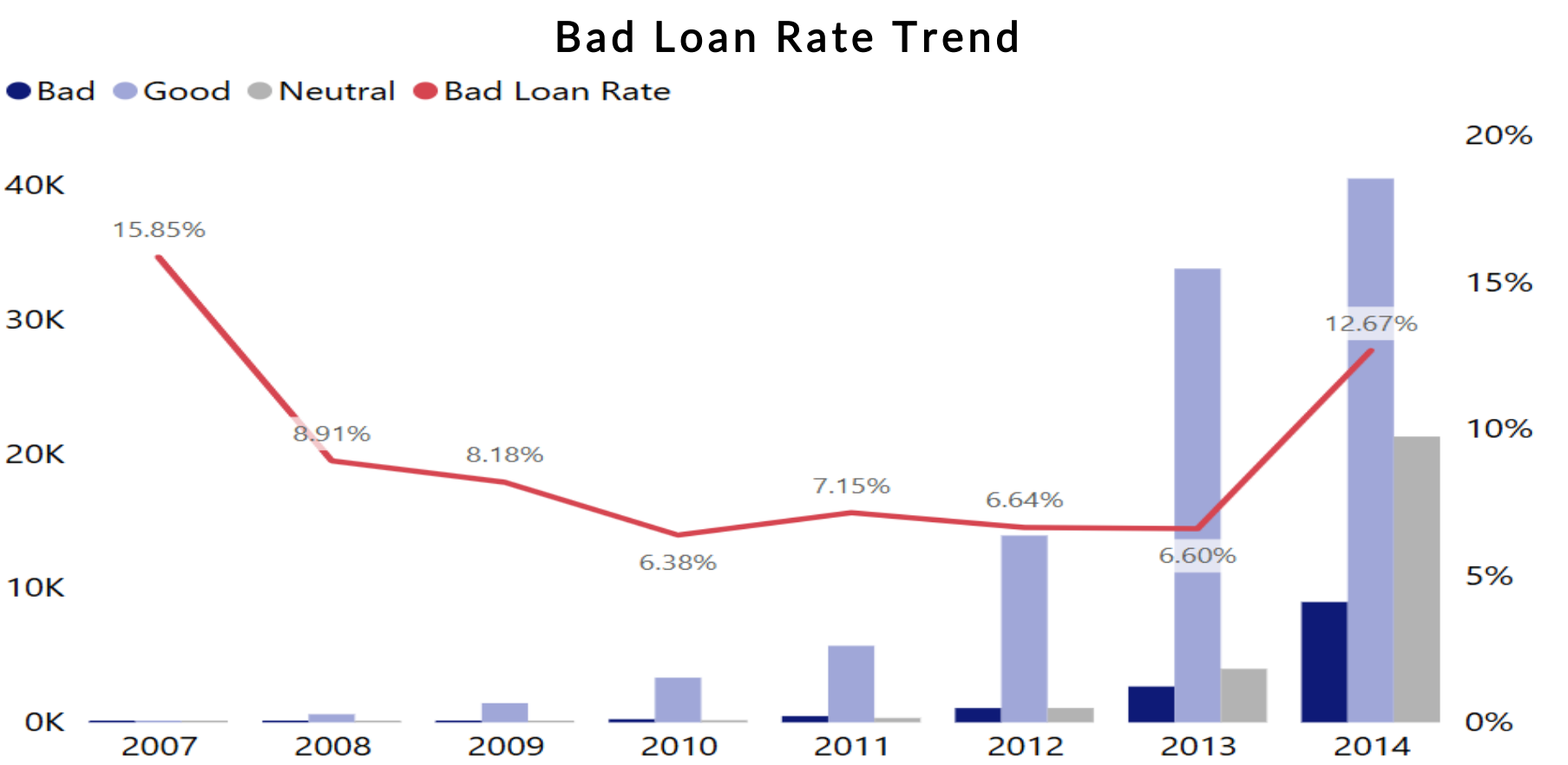
ANALYSIS - Overview - Bad Loan Contribution & Rate Trend

Loan volume grew with a **fluctuating bad rate (6-12%)**, and the **greatest risk** comes from the most **delinquent accounts**.

The **Charged Off** and **Default** statuses together make up 11.7K, which is **over 85%** of the total loan

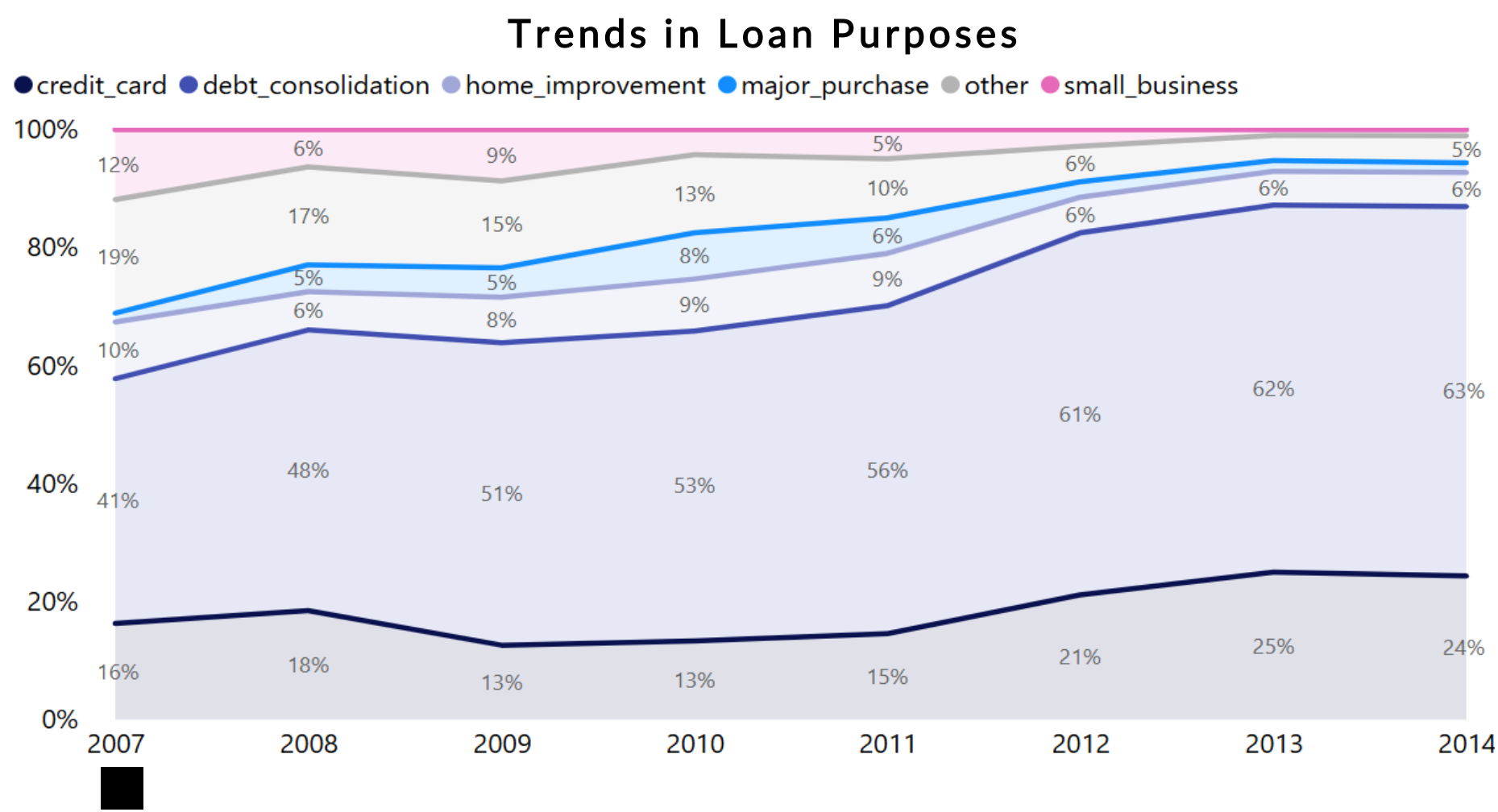


- Loan volume **peaked in 2014** at over 40K,
- The bad rate fell sharply from 2007 to 2010, then stabilized until 2013. In 2014, the rate strongly increased (+6.07%) to **12.67%**.

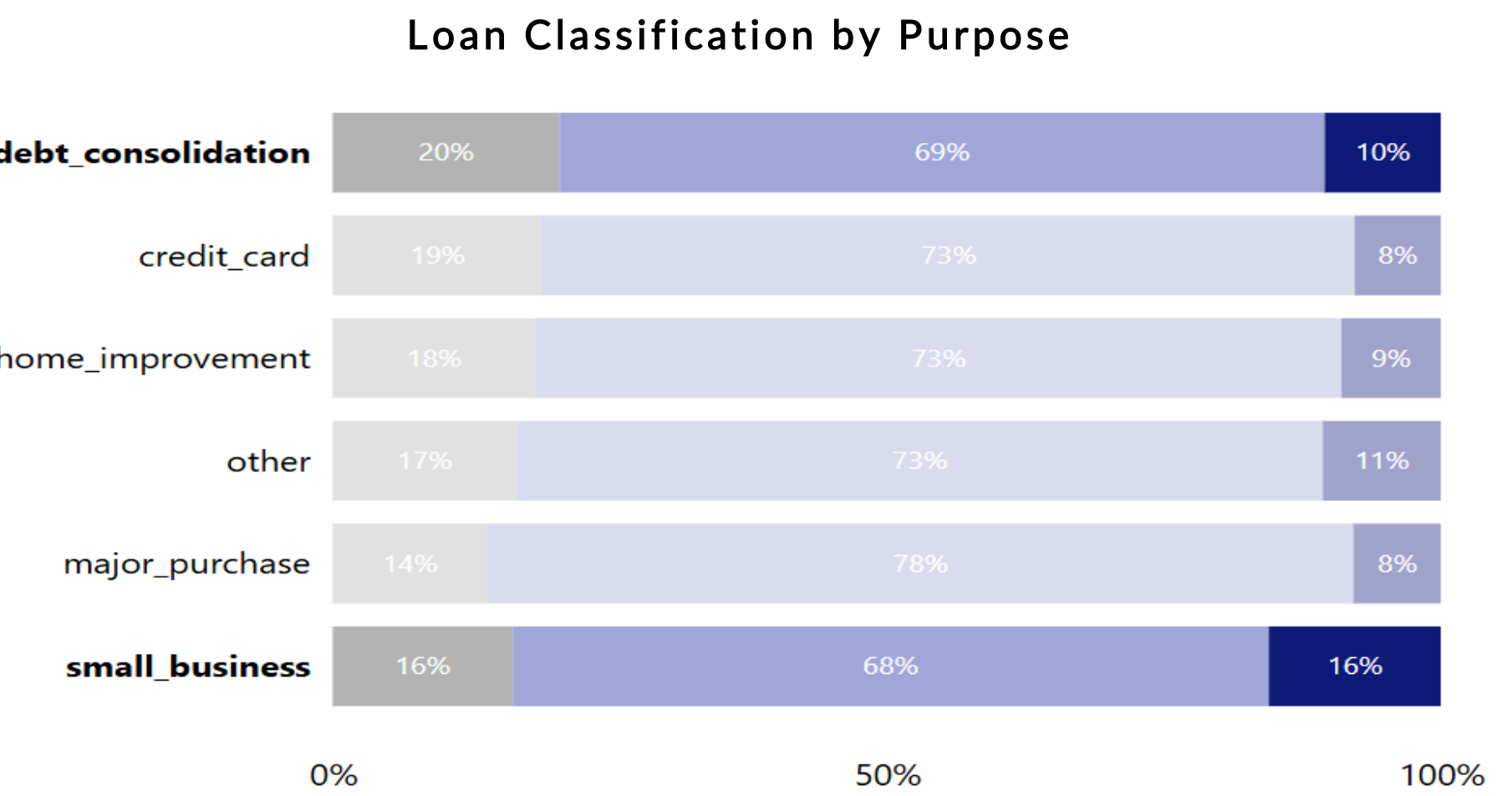


The significant growth of loans for debt-related purposes and Small Business loans is the riskiest.

Debt Consolidation and Credit Cards became the dominant categories, increasing to 63% and 24% respectively, while other purposes declined.



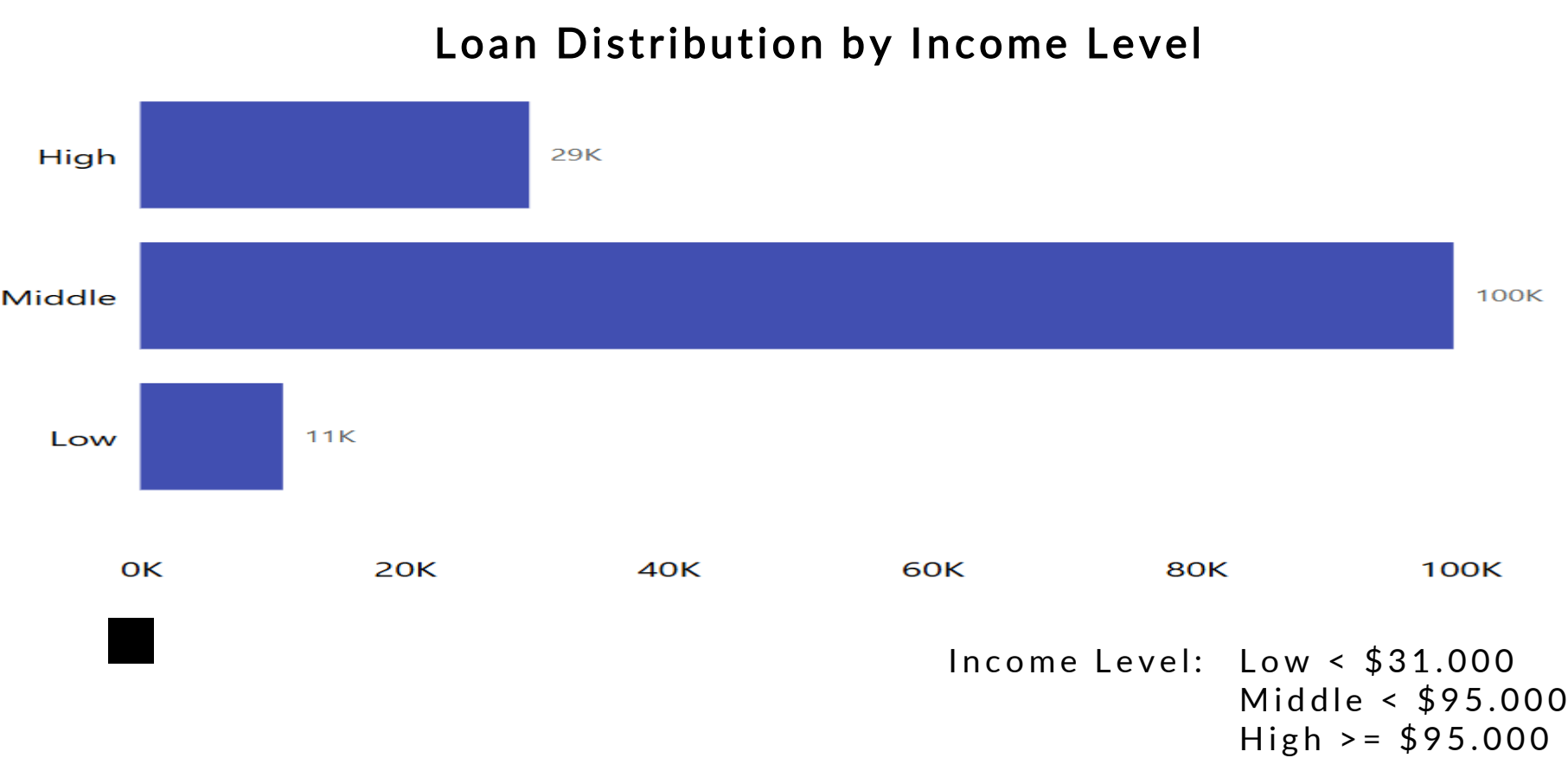
Small Business had the highest percentage of bad loans ~16%, followed by Debt Consolidation loans ~10%



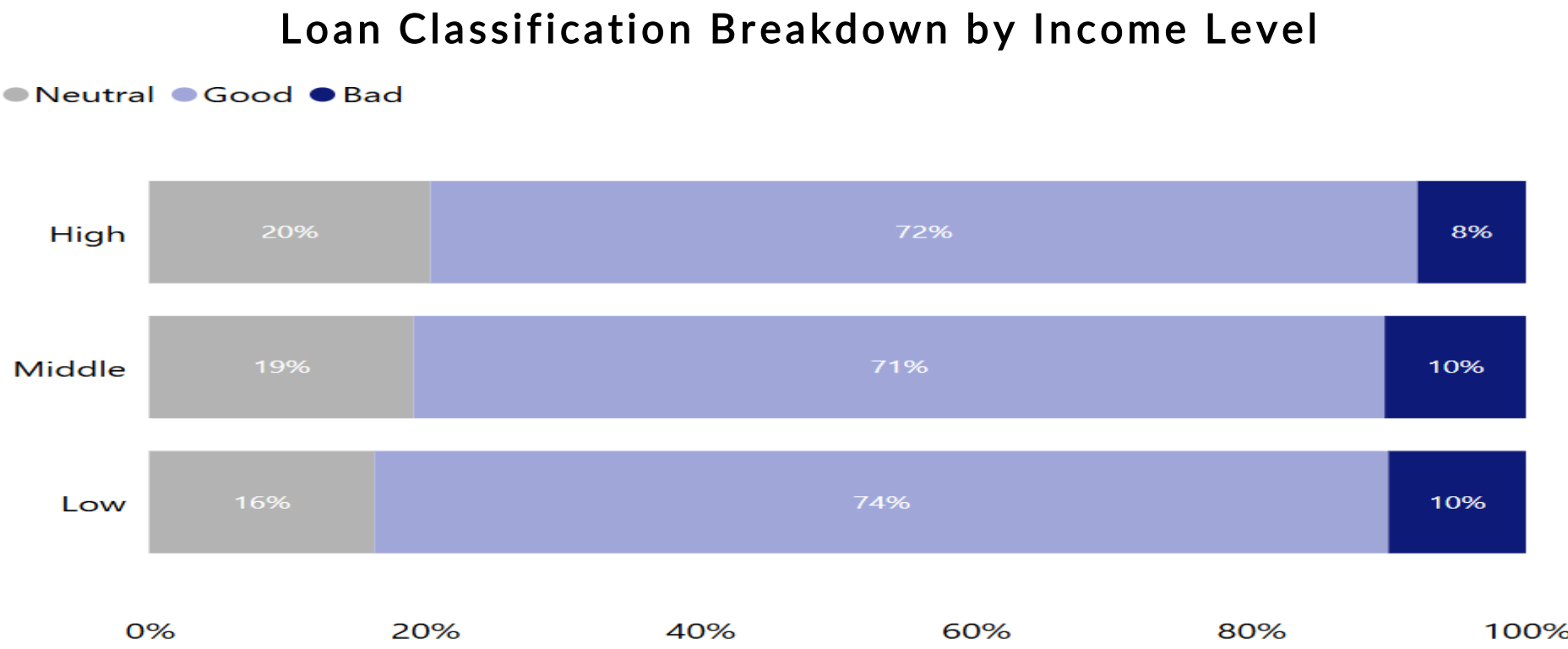


Income’s influence on loan performance. A lower Income shows a higher risk.

Most loans go to the middle-income segment, followed by high-income earners, with low-income individuals receiving the least



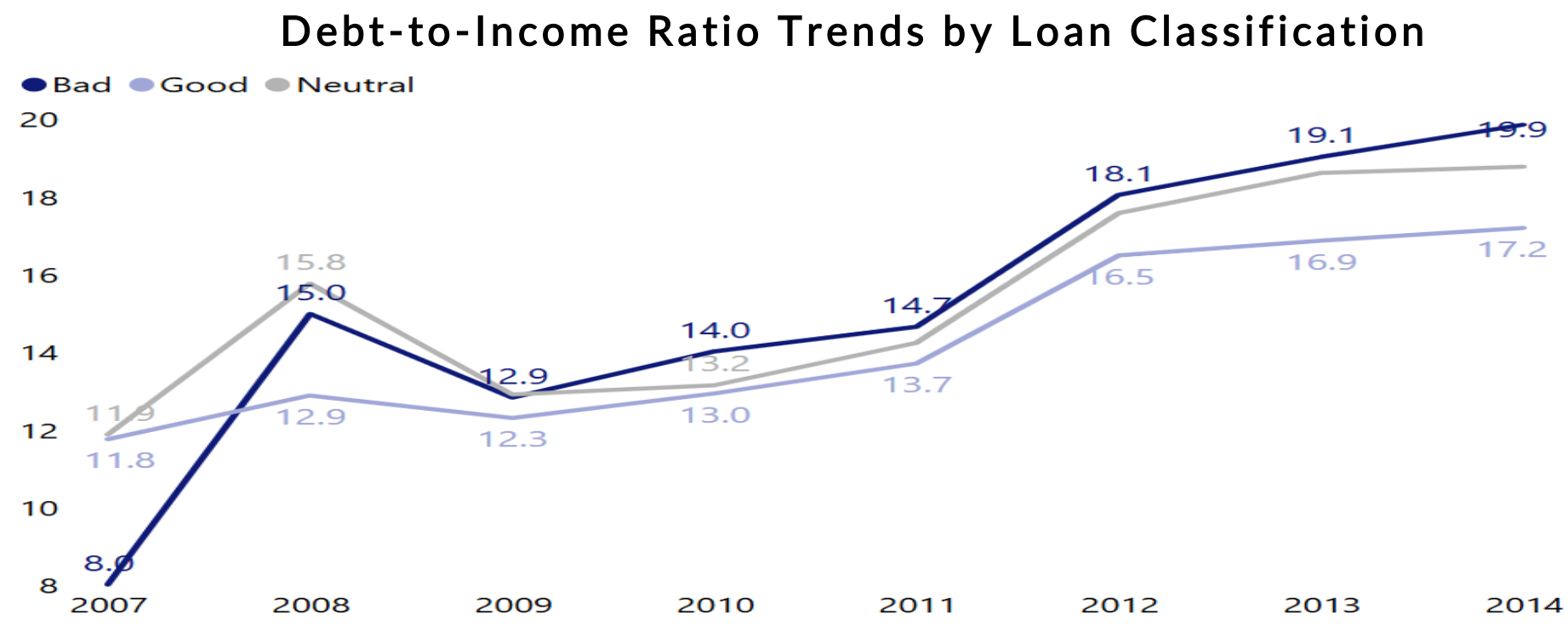
The low and middle-income groups share a similar bad loan rate of 10%; the high-income group has a better performance record with a lower bad loan rate of 8%.



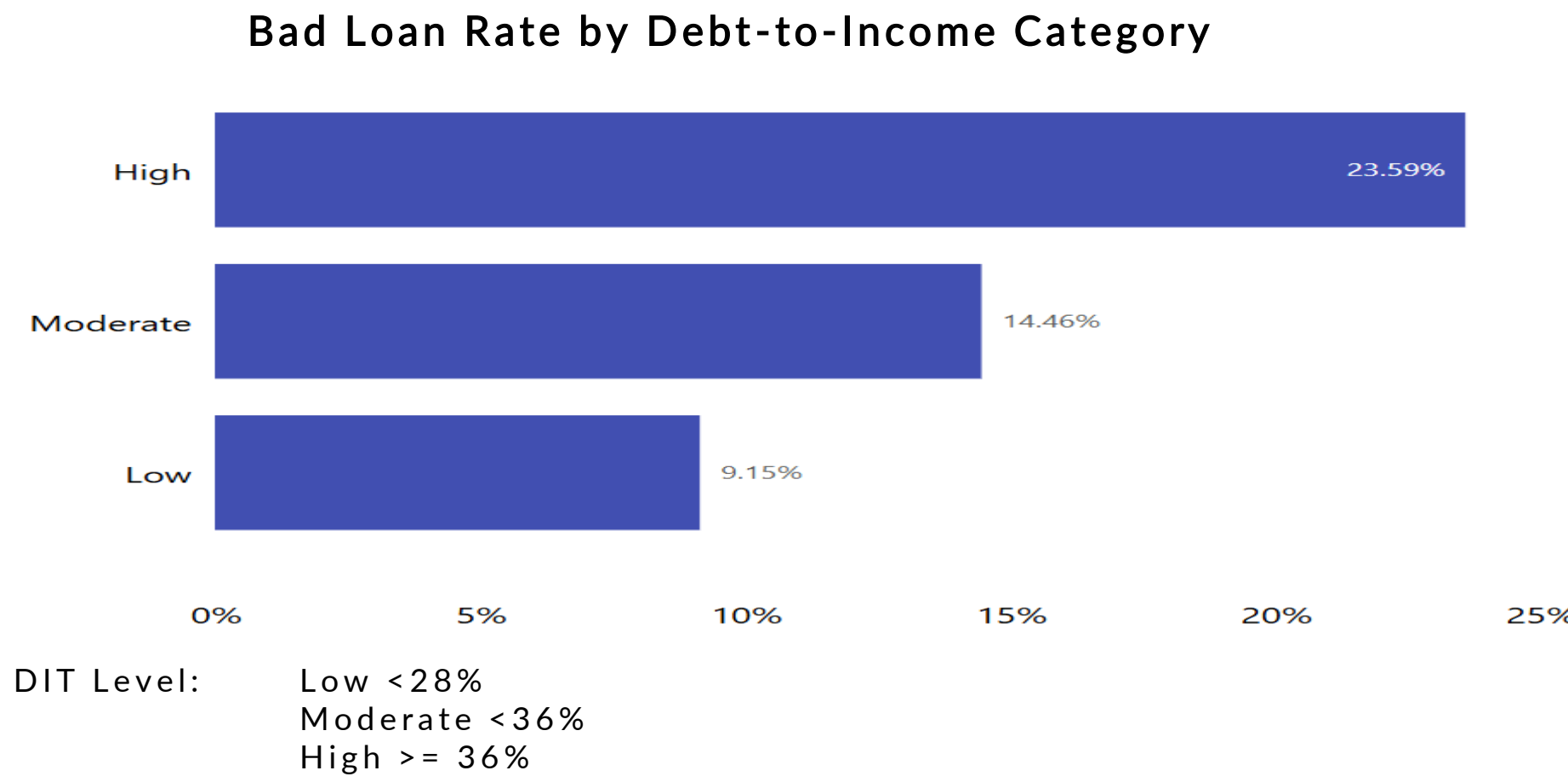


DTI ratios have risen across all loan types, and **higher DTI is linked to a greater risk of default.**

Since 2009, Bad loans have climbed **+7%** from 12.9% to 19.9%
Good loans **+~5%** from 12.3% to 17.2%, and Neutral loans **+~7%**
from 11.9% to 18.8%



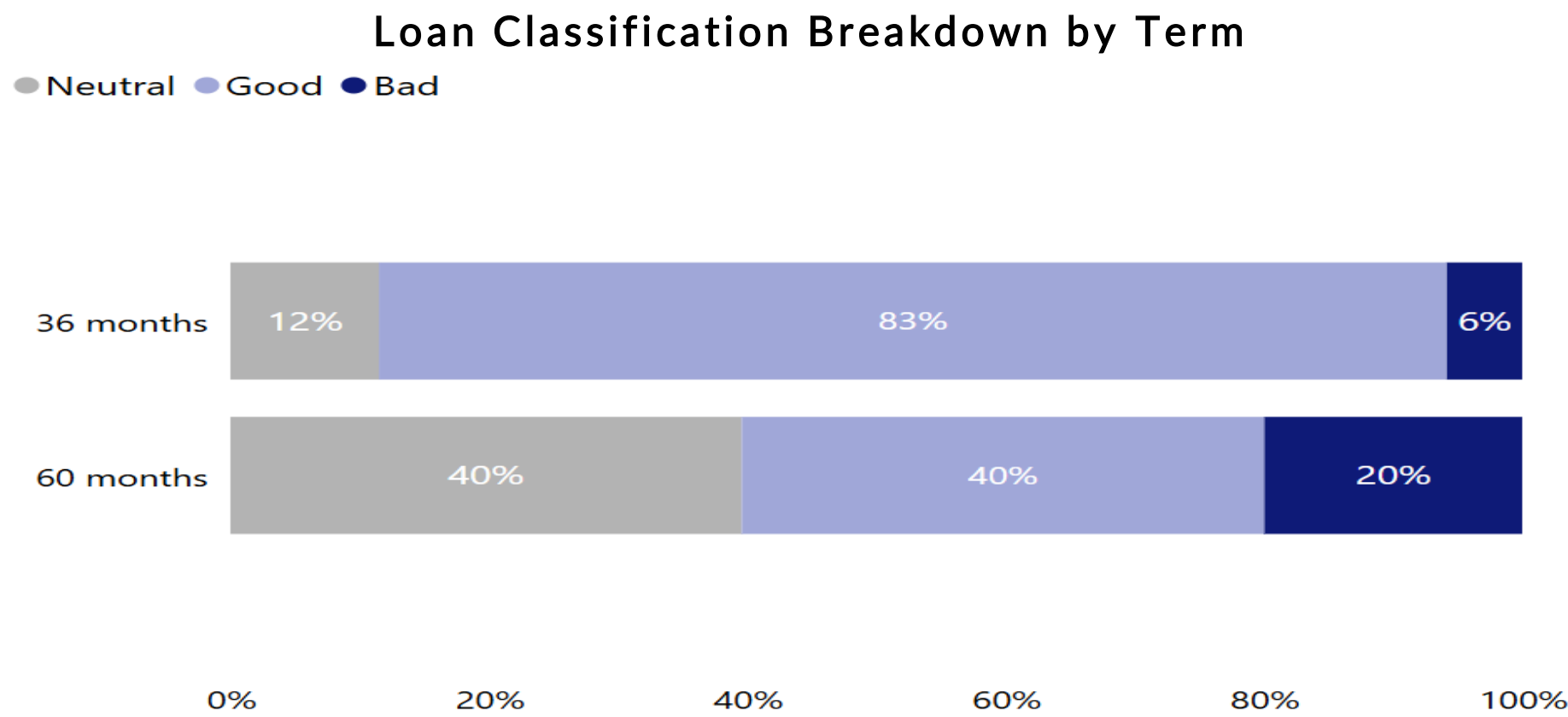
Borrowers with high DTI have a **23.59%** bad loan rate—over
twice the 9.15% rate of low-DTI borrowers



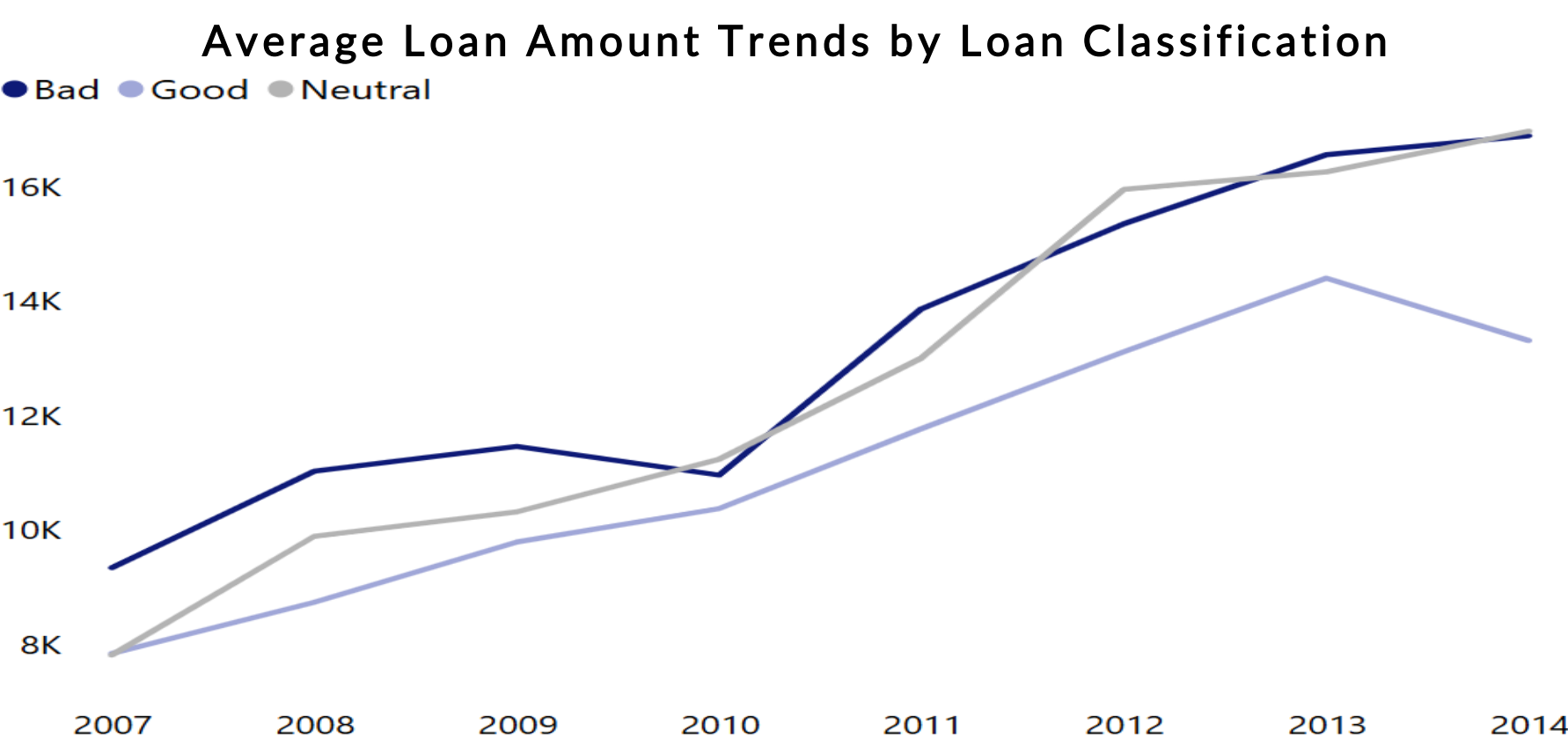
ANALYSIS - Loan Structure - Term & Average Amount

longer terms carry substantially greater risk, and larger loans are associated higher risk

36-month loans have 6% Bad loans and 83% Good loans, whereas 60-month loans rise to 20% Bad loans and 40% Good loans



Since 2010, average loan amounts have risen, with Neutral and Bad loans generally surpassing Good loans from 2011 onward.

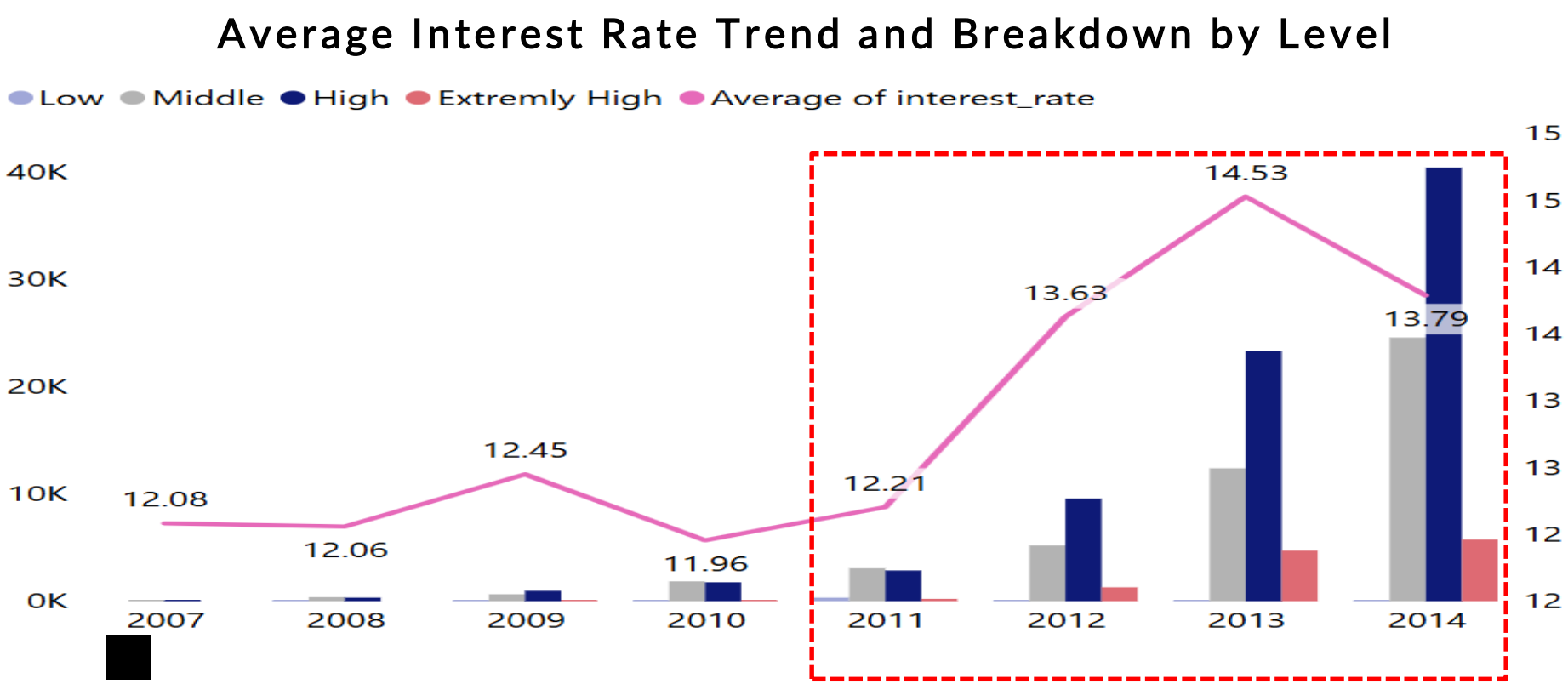


ANALYSIS - Loan Structure - Interest Rate

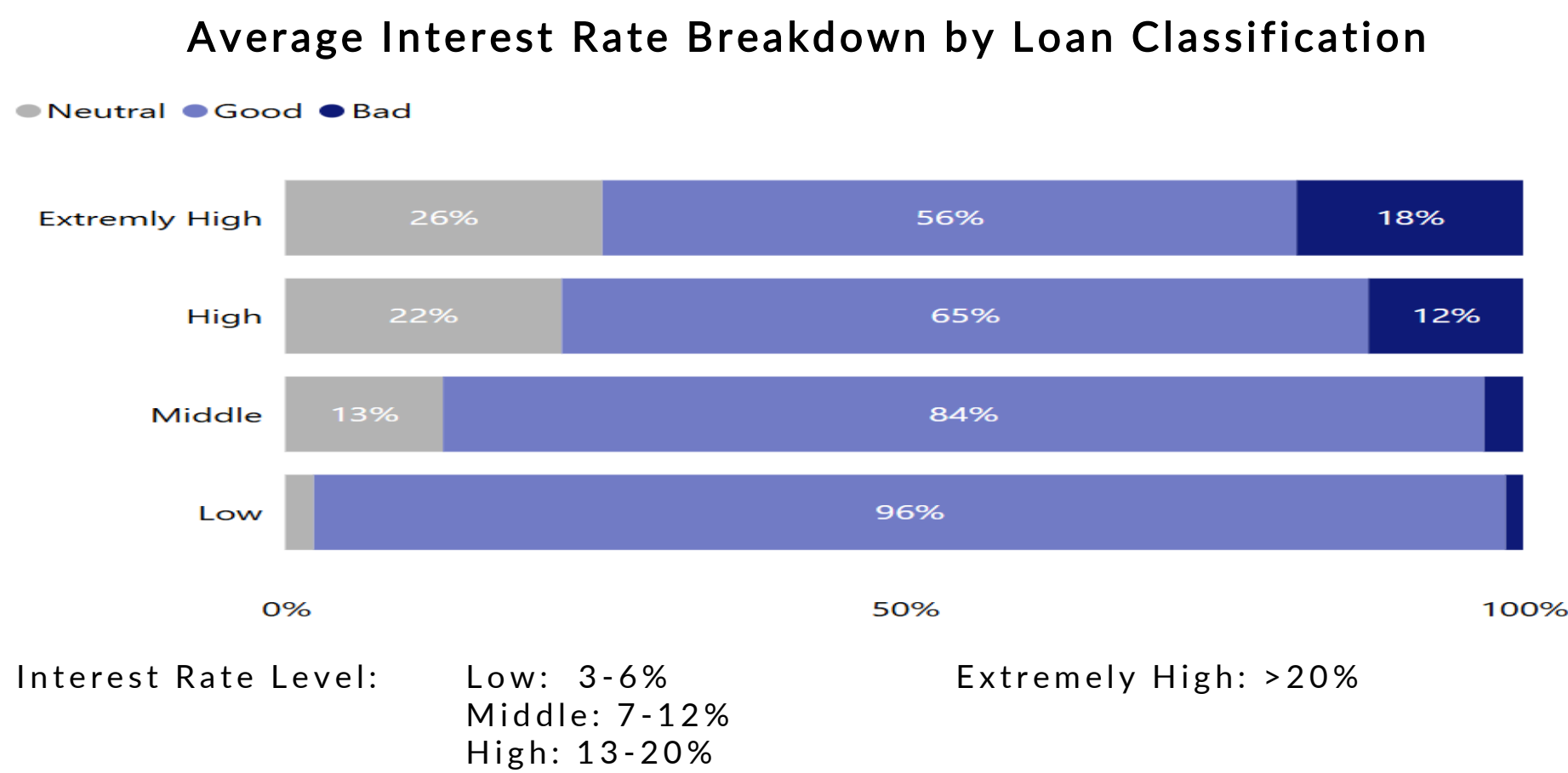
The average interest rate fluctuated between **12-14.5%**. The **higher interest rate** leads to **higher-risk borrowers** with a higher bad loan rate (~18%)

The average interest rate started rising from 2011, reached a peak in 2013 (**14.53%**), and a slight decrease of ~1% in 2014 (13.79%)

Loans concentrated in High and Middle interest rate levels, and the Extremely High interest rate loans had notable growth since 2012.



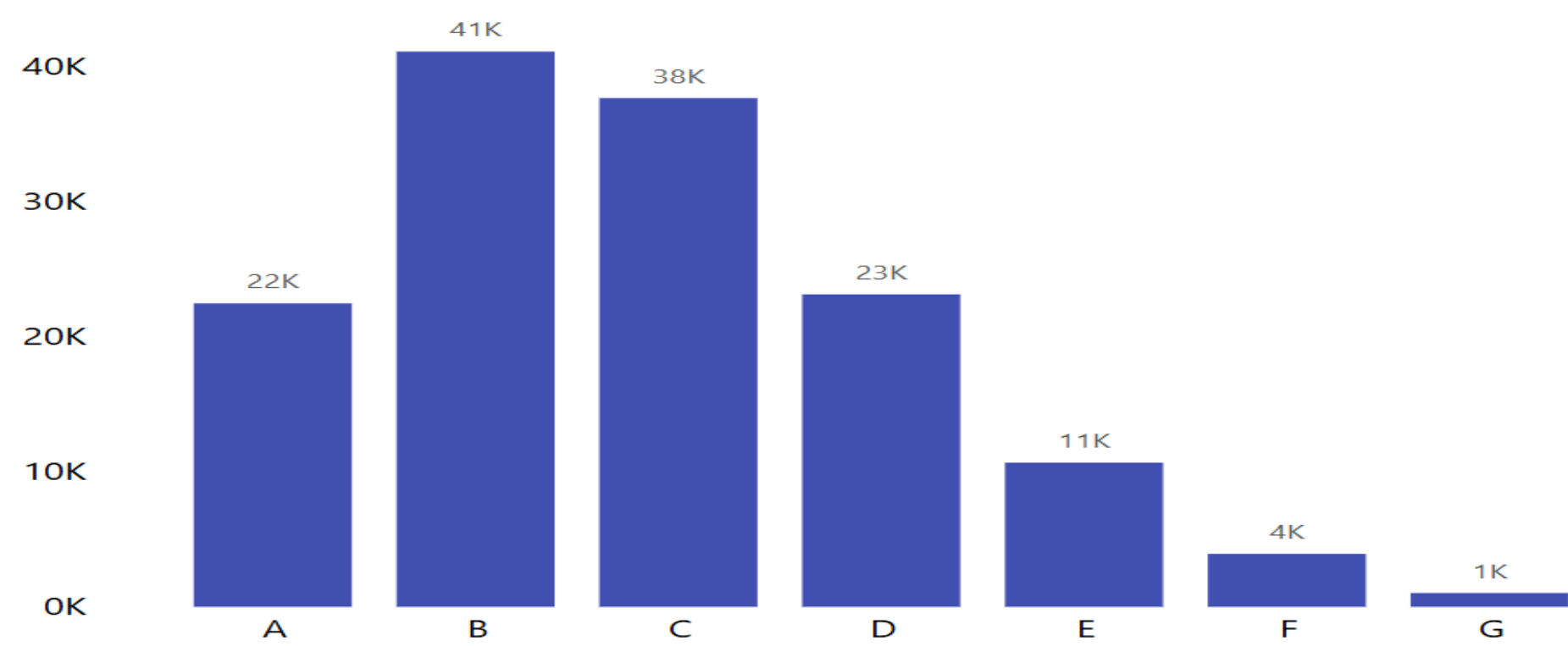
Bad loan rates rise sharply with higher interest, ~0.5% at low rates, 2% at medium, 12% at high, and **18%** at extremely high levels.



Grades **D-G** have the most bad loans, signaling **high default risk**

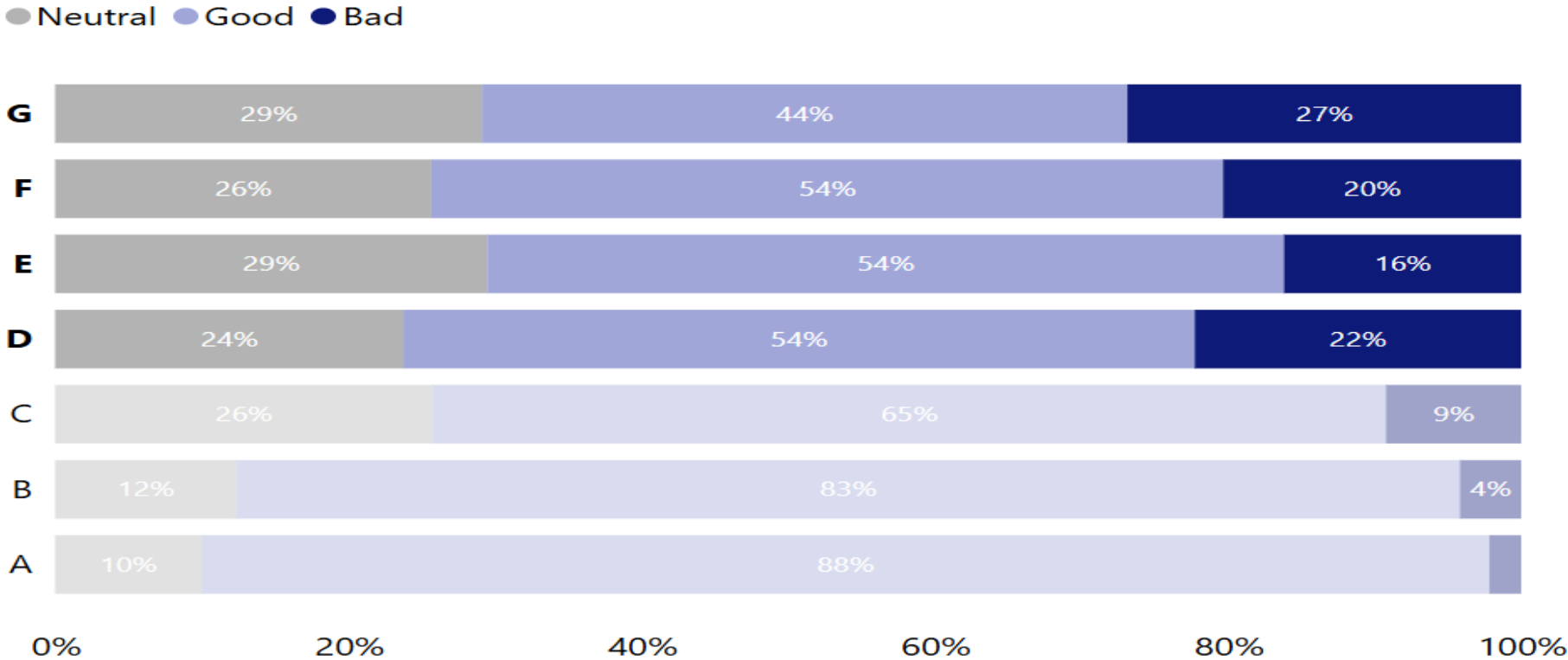
Lending targets average-credit customers, with most loans at B (41k) and C (38k), nearly double A and D; numbers then decline steadily from D (23k) to G (1k).

Loan Distribution by Credit Grade



Bad loans rise sharply as credit grade declines: A (2%) and B (4%) are very low, while **E-G** reach **16-27%**, showing higher default risk.

Loan Distribution Across Credit Grades and Loan Classification



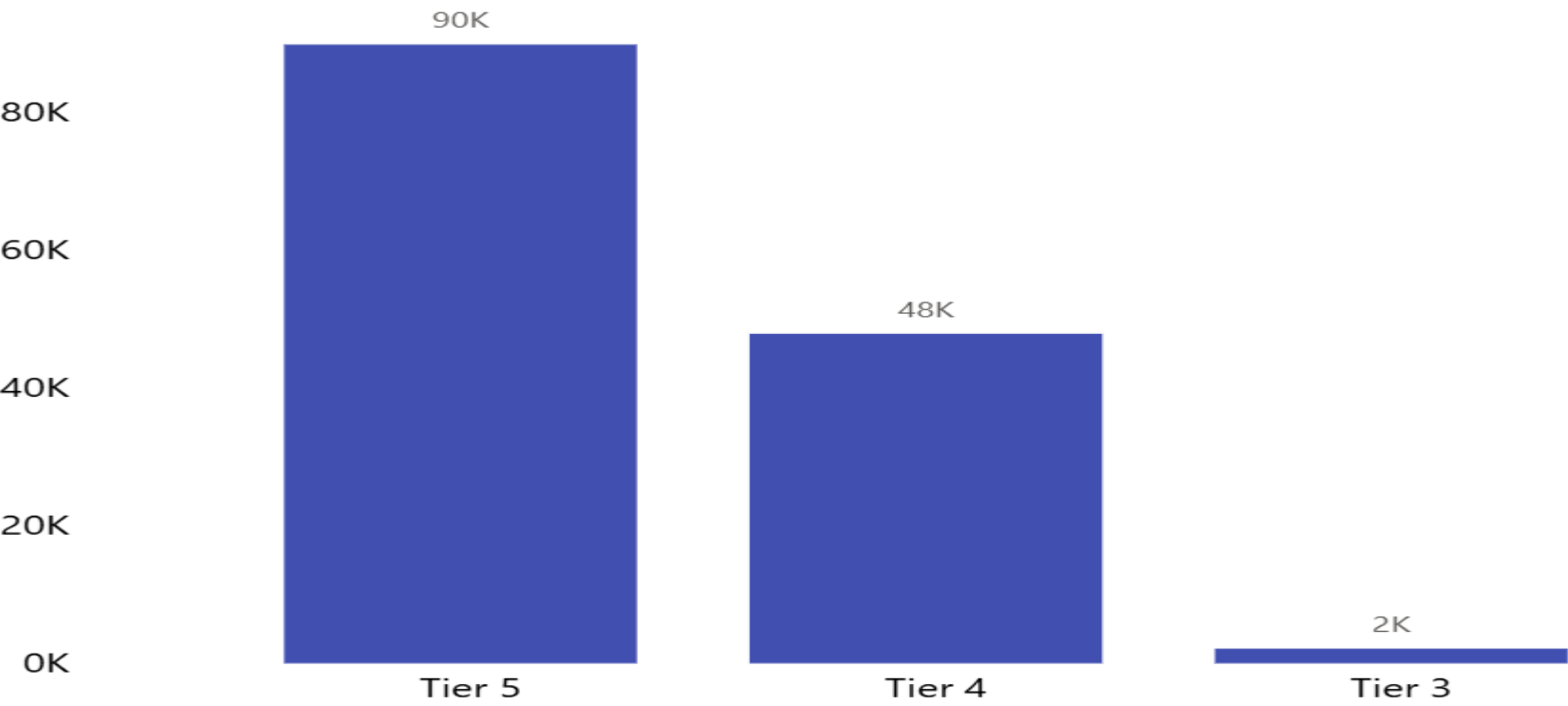
Interest Rate Level: Low <= 6%
Middle <= 12%
High <= 20%
Extremely High > 20



Borrowers with longer credit histories are more active and lower-risk.

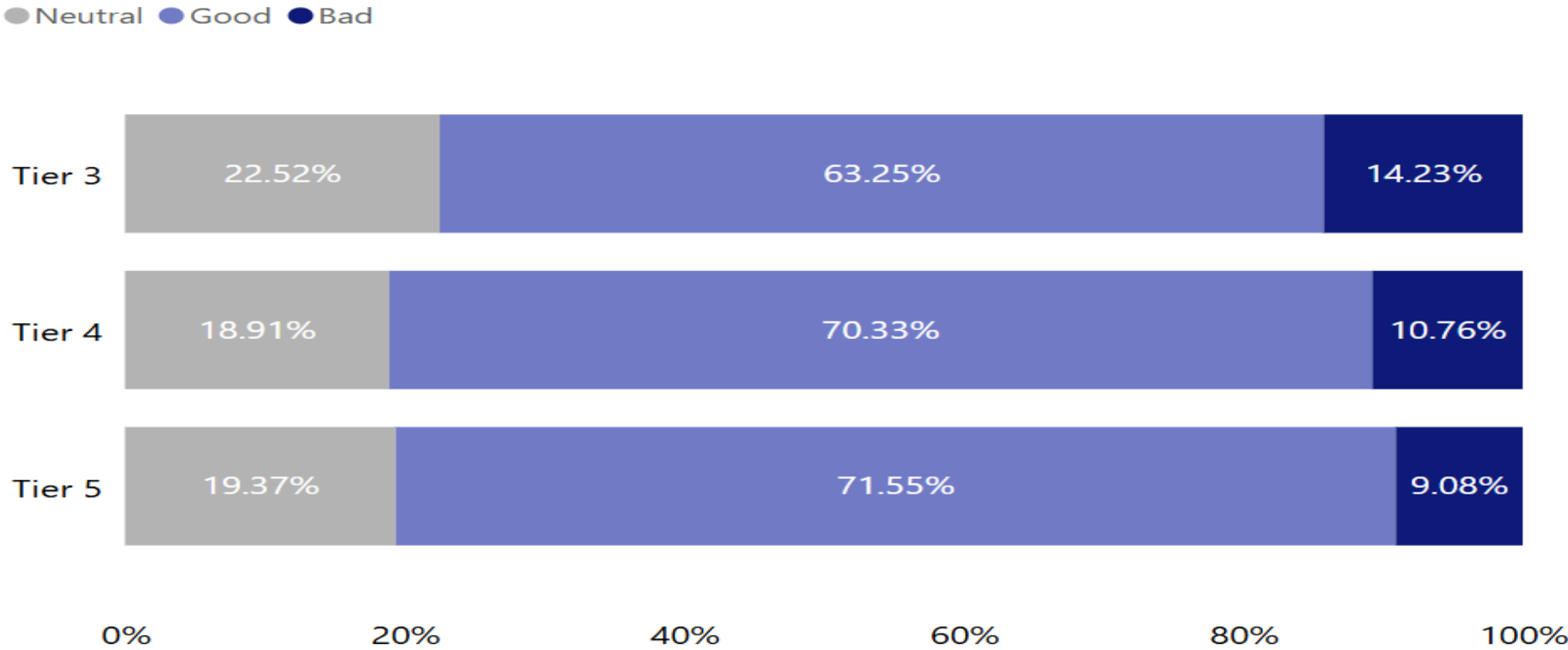
The majority of loan volume (~68%) comes from customers in Tier 5 with a long credit history, which is more than twice the contribution from Tier 4 customers with an established credit history (~30%)

Loan Distribution by Credit Tenure



Good loans increase with longer credit history, from 63.3% (Tier 3) to 71.6% (Tier 5), while Bad loans decrease from 14.2% to 9.1%.

Loan Breakdown by Credit Tenure and Loan Classification

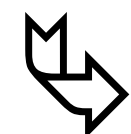


Length Credit History: Tier 1: <1 year
Tier 2: 1-3 years
Tier 3: 3-7 years
Tier 4: 7-15 years
Tier 5: >15 years



KEY RISK FACTORS

- Delinquent accounts
- Debt-related and Small business loans
- Low income / high Debt-to-income ratio
- Long/large loans
- High interest rates
- Grades D–G / short credit histories

 How to minimize risk?



RECOMMENDATION



Pre-lending



Low Risk



Medium Risk



Extremely Risk

Safety Net



Lending



Medium Risk



Low Risk



Post-lending



Bad Loans

Risk Assessment. Identify extremely risky borrowers

- Customer Assessment (KYC & Anti-Money Laundering)
- Creditworthiness Evaluation
- Loan Suitability Assessment
- Behavioral & Transaction Analysis
- Early Risk Indicator Detection
- Decision Rules & Risk-Based Approval

Risk-Based Structuring

- Loan Pricing & Structuring (fit to risk level)
- Credit Limit & Repayment Planning
- Ongoing Monitoring Setup
- Customer Engagement & Education

Risk Monitoring & Control

- Repayment Monitoring & Early Warning
- Customer Engagement & Reminder Systems
- Loan Restructuring & Collection Strategies
- Portfolio Review & Diversification





THANK YOU

30th August, 2025